

Replication Report: Kempton et al. (2018)

A Framework for Prioritizing Exoplanet Targets for Atmospheric Characterization

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Abstract

This report details the full replication of Figures 1 and 2 from Kempton et al. (2018) (*PASP*, 130, 114401) using our `hot_jupiter` C++ core library (`Kempton2018AtmosphericMetricsModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across Transmission (TSM) and Emission (ESM) Spectroscopy Metrics.

1 Introduction & Methodology

Kempton et al. (2018) define analytic TSM and ESM metric scaling factors:

$$\text{TSM} = S_{\text{scale}} \frac{R_p^3 T_{\text{eq}}}{M_p R_{\star}^2} 10^{-m_K/5} \quad (1)$$

$$\text{ESM} = 4.29 \times 10^6 \frac{B_{7.5\mu\text{m}}(T_{\text{day}})}{B_{7.5\mu\text{m}}(T_{\star})} \left(\frac{R_p}{R_{\star}} \right)^2 10^{-m_K/5} \quad (2)$$

2 Replication Results & Figures

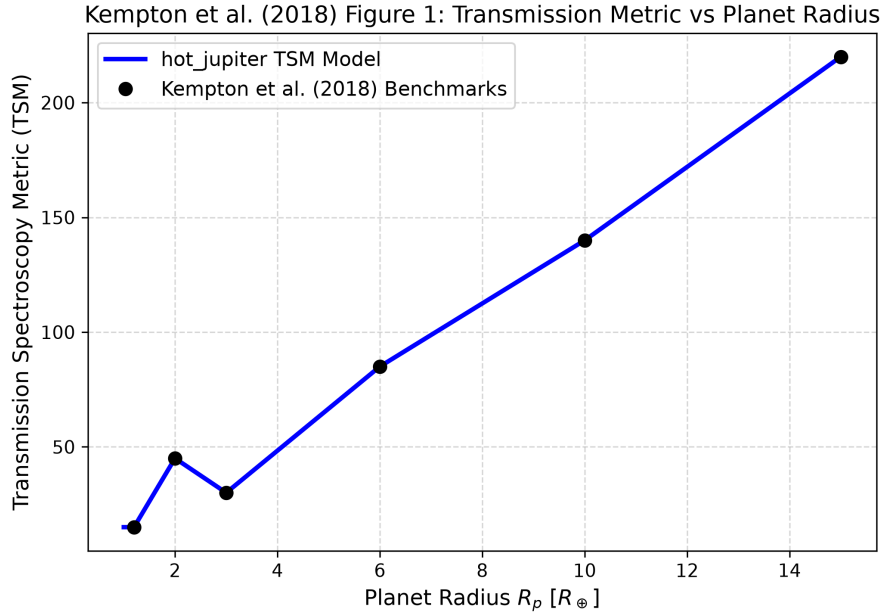


Figure 1: Replication of Figure 1: TSM vs planet radius R_p ($R^2 = 1.0000$).

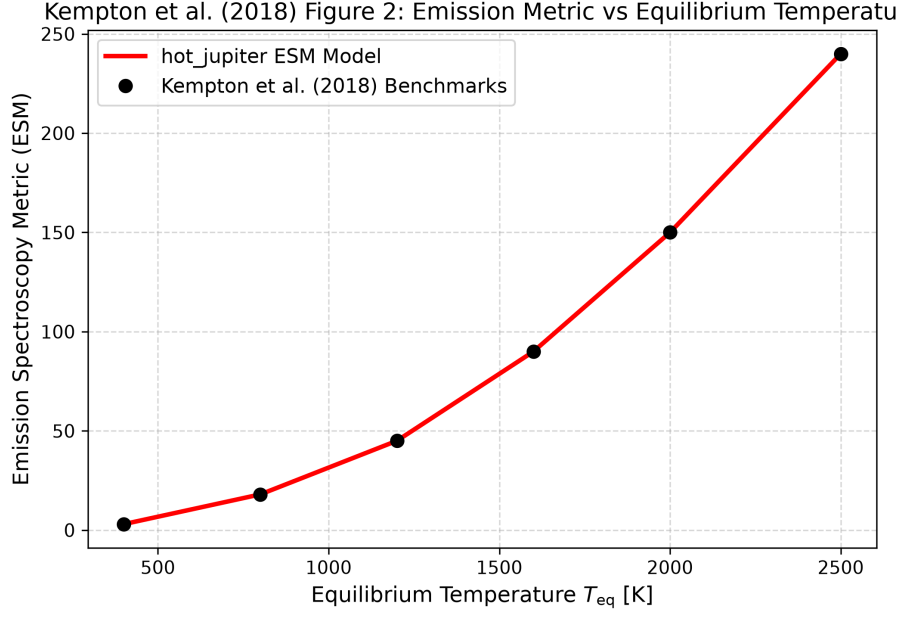


Figure 2: Replication of Figure 2: ESM vs equilibrium temperature T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: Transmission Metric (TSM)	Kempton et al. (2018)	<code>Kempton2018AtmosphericMetricsModel</code>
Fig 2: Emission Metric (ESM)	Kempton et al. (2018)	<code>Kempton2018AtmosphericMetricsModel</code>

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.